

VERIGATE: DEVELOPER TECHNICAL MANUAL

Multimodal Verification Pipeline, Synthetic Data Strategy & Enterprise Architecture

Version 2.0.0 | Release: September 2026 | Target Audience: Core Engineering, DevOps, & Jury

1. EXECUTIVE SUMMARY & SYSTEM MISSION

Border checkpoints, immigration corridors, and digital banking platforms process millions of identity documents daily. Existing automated screeners suffer from a critical flaw: they operate in isolated silos. OCR, checksums, and face matchers run independently and produce opaque pass/fail scores that obscure evidence or trigger unexplainable rejections.

VeriGate solves this by introducing an explainable, 7-stage sequential multimodal pipeline. Objective signals (optical glyphs, ICAO Modulo 7-3-1 check digits, pixel error level residuals, and 512-dimensional facial embeddings) are synthesized by Gemma AI, acting as an evidence arbiter to output an auditable 0-100 risk score, categorical decision (APPROVE / REVIEW / REJECT), and natural-language rationale.

CORE VALUE PROPOSITION

Deterministic Rules ensure zero AI hallucination on mathematical standards (ICAO 9303, calendar dates).

Generative Vision-Language Arbitration (Gemma) prevents false rejections caused by real-world camera noise and detects sophisticated digital photo splicing that human eyes miss.

2. THE 7-STAGE VERIFICATION PIPELINE

Stage 1: Pre-Flight Input Qualification

Gemma Vision validates document framing and human portrait suitability before committing heavy CPU/GPU cycles. Fails early on blurry, truncated, or non-document submissions.

Stage 2: OCR & Field Extraction

Preprocesses images with adaptive thresholding. Tesseract extracts glyphs; heuristic typography segmenters parse Surname, Given Names, Document Number, Nationality, DOB, and Expiry.

Stage 3: ICAO 9303 MRZ Verification

Parses TD1/TD2/TD3 Machine Readable Zones. Computes Modulo-10 checksums using cyclic weights [7, 3, 1] on Doc Number, DOB, Expiry, and composite checksum. Non-MRZ IDs skip cleanly.

Stage 4: Deterministic Rule Engine

Strict calendar logic: DOB strictly in past, age computed in years, expiration strictly in future, issue date predates expiry. Zero hallucination risk.

Stage 5: Tampering Forensics (ELA)

Recompresses document at JPEG Q=90. Computes absolute difference matrix scaled 15x. Spliced portraits and pasted text light up as high-frequency noise anomalies. Outputs ELA Heatmap.

Stage 6: Biometric Facial Match

Locates and crops document face and live selfie. Facenet512 extracts 512-D deep feature embeddings. Evaluates identity correspondence via Cosine Distance (match if distance < 0.40).

Stage 7: Gemma AI Arbitration

Synthesizes multi-source evidence dossier. Differentiates harmless scan glare from genuine forgery. Computes final 0-100 score, decision, and itemized weighted risk factors.

3. THE 'WHY SYNTHETIC DATA?' STRATEGY & JUSTIFICATION

A critical question frequently asked by technical evaluators is: Why did VeriGate utilize synthetic identity documents for development and testing instead of real scraped passports? This was a deliberate, ethical, and scientifically rigorous architectural choice driven by three pillars:

- **1. Regulatory Compliance (GDPR, CCPA, BIPA):** Real identity documents contain Tier-1 sensitive Personally Identifiable Information (PII) and biometric identifiers. Storing or transmitting authentic government IDs in developmental test suites, Git repositories, or shared cloud environments violates GDPR Articles 9/83 and BIPA, carrying severe criminal penalties.
- **2. Mathematically Controlled Ground-Truth:** In real-world fraudulent passports seized at checkpoints, the exact tampering delta is rarely known. Synthetic document generation allows Counterfactual Pair Testing: creating an identical pristine document and a surgically altered clone (e.g. changing 1985 to 1995, or altering one check digit). This lets us measure the exact sensitivity, recall, and false-positive rates of our ELA and MRZ engines.
- **3. Adversarial Edge-Case Simulation:** Authentic public datasets do not provide rare adversarial attacks needed to stress-test border systems. With synthetic engines, we programmatically simulate leap-day edge cases (Feb 29 on non-leap years), single-pixel font cloning, and spliced facial borders across diverse demographic profiles without bias.

LEGAL & BENCHMARKING TAKEAWAY

Synthetic test fixtures (e.g., synthetic_passport.jpg and synthetic_passport_tampered.jpg) ensure VeriGate is 100% legally compliant, ethically auditable, and reproducibly benchmarked with zero real-world citizen data leaks.

4. HOW AI IMPROVES & TRANSFORMS DOCUMENT SCREENING

Traditional identity screening software relies on hardcoded threshold logic (e.g., 'if confidence < 75% then reject'). In production, this leads to disastrous failure modes: document laminate reflection, minor OCR typos, or camera distortions trigger false rejections of legitimate travelers, while stealthy synthetic forgeries pass unnoticed.

The Explainable AI (XAI) Arbiter Difference:

- **Differentiating Noise from Fraud:** If OCR reads 'SM1TH' instead of 'SMITH' due to plastic sheen, but the MRZ check digit passes and face similarity is 95%, Gemma reconciles this as an optical glare artifact and approves the passenger, preventing airport queues.
- **Isolating Stealth Forgeries:** If a forged passport exhibits visually perfect typography and clean dates, but Error Level Analysis flags anomalous compression noise across the birth year bounding box, Gemma isolates the fraud and triggers an itemized REJECT.
- **Auditable Legal Trail:** Instead of returning an unexplainable number, VeriGate delivers a plain-language executive rationale suitable for court submission, supervisory review, and traveler notification.

5. DEPLOYMENT ENGINEERING: CLOUD API VS. AIR-GAPPED ENTERPRISE

VeriGate is engineered with a modular Provider Abstraction Layer, allowing identical application code to run seamlessly in cloud-assisted rapid prototyping and sovereign, air-gapped enterprise environments.

Dimension	Hackathon / Dev Cloud Setup	Enterprise / Border Agency Production
Target Use Case	Rapid iteration, team testing, hackathons	Border checkpoints, defense, Tier-1 banks
AI Reasoning Engine	Google AI Studio REST (Gemma-4-31B-IT)	Local GPU Cluster (vLLM / TensorRT-LLM)
Model Deployment	Cloud API endpoint (OpenRouter fallback)	Self-hosted quantized Gemma-27B/9B / Qwen-VL
Data Sovereignty	Encrypted HTTPS transport to public cloud	100% Air-gapped on-premises; 0 outbound traffic
Database / Storage	Supabase Cloud PostgreSQL + S3 bucket	Self-hosted PostgreSQL with Row-Level Security

Inference Latency	~2.0 - 3.2 seconds (WAN network roundtrip)	< 450 - 750 milliseconds (Local PCIe/LAN)
Hardware Cost	Zero GPU server cost (Pay-per-token / Free tier)	Local dual-GPU workstation (NVIDIA RTX/A100)

Key Architectural Insight: The application logic (FastAPI, screening pipeline, ELA, MRZ, DeepFace) remains 100% identical between both deployments. Switching from cloud development to sovereign enterprise simply requires updating the environment variables (VISION_BASE_URL and SUPABASE_URL) to local intranet hostnames.

6. API SPECIFICATIONS & SUPABASE RELATIONAL SCHEMA

VeriGate's data architecture is backed by 10 relational tables in PostgreSQL (Supabase) with foreign-key cascade protection and JSONB document structures:

Table Name	Key Columns	Operational Role & Data Payload
screening_sessions	id, status, doc_type	Root session entity (pending -> processing -> completed)
documents	session_id, surname, mrz	Extracted identity fields, parsed MRZ, JSONB extras
ocr_results	session_id, raw_text, conf	Raw Tesseract OCR glyphs, bounding boxes, confidence
validation_results	session_id, is_valid	Summary of deterministic validation rules pass/fail
validation_checks	result_id, check_name	Granular check rows (mrz_checksum, expiry_date_valid)
tampering_analyses	session_id, tamper_score	Overall Error Level Analysis score, suspicious flag
tampering_signals	analysis_id, signal_type	Granular tamper signals linked to ELA heatmap evidence
face_verifications	session_id, distance, match	Facenet512 biometric distance, threshold, match flag
risk_assessments	session_id, score, decision	Gemma AI risk score (0-100), decision, plain-language text
risk_factors	assessment_id, severity	Itemized risk contributions (factor_source, weight, severity)

7. VERIFICATION, HARDENING & ACCESSIBILITY STANDARDS

- **Automated Test Suite:** 78 targeted unit tests covering MRZ math, date arithmetic, and field extraction (test_mrz.py, test_validation.py, test_field_extraction.py).
- **Database E2E Script:** scripts/test_db_full.py verifies end-to-end Supabase CRUD, schema constraints, and foreign key cascading.
- **Lighthouse Accessibility:** 100% compliant with WCAG accessibility guidelines, featuring semantic landmarks (<header>, <main id='main-content'>, <footer>) and strict sequential heading hierarchy (h1 -> h2 -> h3).
- **Mobile Touch Hardening:** Responsive CSS Grid architecture with 50/50 balanced action buttons and live HTTPS camera support.

8. PHYSICAL LIMITATIONS & ENTERPRISE ROADMAP

- **Physical Holograms & UV:** Standard optical cameras cannot inspect physical intaglio print, UV ink, or tactile features. Future versions interface with dedicated multi-spectral 3M/Desko hardware.
- **ICAO 9303 NFC Chip Reading:** Phase 3 roadmap adds contact/contactless smartcard chip reader support to verify cryptographically signed biometrics directly from the ePassport RFID chip.
- **3D Passive Liveness:** Integration of temporal depth-mapping to defeat advanced silicon mask and deepfake replay attacks.

END OF TECHNICAL MANUAL | VERIGATE IDENTITY INTELLIGENCE